PBL Report Writing

Guidelines

v1.1

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*This document outlines the general guidelines to be followed by students undertaking the **Project-Based Lab (PBL)** for the Compiler Design course in the Department of Computer Science and Engineering, NIT Warangal. Students are advised to carefully read and adhere to both the general guidelines provided herein and any specific instructions communicated by the course instructor. Strict compliance with these guidelines is essential to ensure uniformity, technical rigor, and high-quality documentation of the project work.*

Structure

Report Writing

Structure of a Report

- Abstract
- Introduction
- Background Information
- Motivation
- State of the Art (Related Work)
- Design of Solution
- Realization of Solution
- Validation\Evaluation of Solution
- Conclusions and Future Work

Abstract/PBL Summary

- Reflects all parts of your work in a shortened form
- Structure
  - General area of the problem.
  - What problem did you study, and why is it important?
  - What methods did you use to study the problem?
  - What are the key results?
  - What is new about them?
  - What are the conclusions and perspectives based on the results?
- Often poorly written (e.g., lacks important information, presents biased information)
- Avoid a copy-paste (from the introduction) approach, i.e., write a unique and effective summary
- Do not cite references

Introduction

- Context of research (need and importance)
- Brief overview of the state of the art
- Gaps in the state of the art
- Research question
- Methodology
- Your original contribution (Not AI Tools Content)
- Contents of the PBL

Background Information

- Theories and concepts that may be unfamiliar to the instructor and are essential for understanding the remainder of the PBL
- Should be focused
- Should be critical, consistent, and logically structured

State of the Art (Related Work)

- Identify and describe the key research areas related to your topic
- Summarise the current state of the art in each area, highlighting major works and developments
- For each related work:
  - Explain what it is and its main contribution
  - Describe how it builds on or differs from previous approaches
  - Discuss its limitations or research gaps that remain
- Position your own work within this context; show how it addresses identified gaps or advances the field
- Ensure your writing is clear, focused, and well-organized

Design of Solution/Architectural Design

- It represents the abstract design of the solution
- It can be an architectural design, an algorithm, or another form
- The title of this section depends on the content

Realization of Solution/Methodology

- It provides implementation details
  - How is the design practically realized?
- The title of this section depends on the content

Validation/Evaluation of Solution

Mathematical proof (If Used)

- Axioms
- Lemma
- Theorem
- Corollary
- Conjecture

Statistical validation, use cases, case study

- Evaluation metrics
- Experimental setup
- Execution
- Data representation
- Data analysis
- Data interpretation

Conclusion and Future Work

- State the most important outcome of your work
  - Interpret your findings
- Show whether (or to what extent) you have succeeded in addressing the PBL question
- Include perspectives (Future Work)
  - What can still be done in relation to the PBL question

Language

Use formal, simple and concise writing

Language

- Use active voice whenever possible (check the venue about this, some prefer passive)
  - Active: *We <u>present</u> an approach to automation.*
  - Passive: *An approach to automation <u>is presented</u>.*
- When to use passive voice [1]:
  - To emphasise the action rather than the actor
    - *After long debate, the proposal was endorsed by the long-range planning committee.*
  - To keep the subject and focus consistent throughout a passage
    - *The data processing department recently presented what proved to be a controversial proposal to expand its staff. After long debate, the proposal was endorsed by the long-range planning committee.*
  - To be tactful by not naming the actor
    - *The procedures were somehow misinterpreted.*
  - To describe a condition in which the actor is unknown or unimportant
    - *Every year, thousands of people are diagnosed as having cancer.*

Language

- Use the present tense whenever possible
- User short sentences
- Text flow must be guaranteed
  - No breaks in the text between two paragraphs, sections, etc.
- Avoid opening the paper, section, or paragraph with *This* or *In this*
- Connect section titles, figures, tables, and other ancillary elements to the main text by referencing
  - When referencing them, capitalize the references
    - “Table 1”, “Section 2”, “Figure 3” or “Fig. 3”, “Algorithm 1”, etc.

Language

- Spelling and punctuation should be perfect
- Be consistent
  - Spelling of technical terms, use of terminology, acronyms, etc.
  - For example, avoid using synonyms for technical terms
    - If several terms are available, choose one and use it consistently
- Correct spelling of established terms
  - For example, "**Compiler Design**", etc.
- Do not use contractions

Language

- Avoid noun strings
- Use pronoun references carefully
  - All pronouns must clearly refer to definite nouns
  - Unclear: *Alan Turing was an excellent mathematician and logician. This is how he investigated machine intelligence.*
  - Clear: *Alan Turing, who was an excellent mathematician and logician, used his ability with numbers and logic to investigate machine intelligence.*

Referencing

Referencing

- Plagiarism is not allowed
- Plagiarism is claiming someone else's work as your own
- Plagiarism can be avoided by using quotations and referencing
- If you copy a text from somewhere and paste it into your paper, you must
  - Quote it (short quotations in double quotes and longer quotations in block quotes) <u>and</u>
  - Provide a reference for it
- If you paraphrase a text, you must provide a reference for it
- If you copy an illustration from another source, you must provide a reference for it
- Read [2]

Referencing

- Use scientific references
  - Papers in scientific conference proceedings, articles in scientific journals, standard reference books, etc.
  - Limit the use of references to Wikipedia
- When using a link in or as a reference, the date when the Webpage was last visited must be specified
  - Option 1: specify a separate date in each citation. Check the @online entry (and its urldatefield) of the BibLaTeX package.
  - Option 2: Put a sentence about the last time links were checked directly below the bibliography list.

References

- [1] Writing Center, The Writer's Handbook: Use the active voice, University of Wisconsin – Madison, USA. URL: https://writing.wisc.edu/handbook/style/ccs_activevoice/ (visited on 30.09.2020)
- [2] Roig, Miguel, Avoiding plagiarism, self-plagiarism, and other questionable writing practices: A guide to ethical writing. URL: https://ori.hhs.gov/sites/default/files/plagiarism.pdf (visited on 30.09.2020)

Further Reading

- Justin Zobel. 2015. Writing for Computer Science (3rd Ed.). Springer Publishing Company, Incorporated.
- The University of Chicago Press Editorial Staff . 2017.The Chicago Manual of Style (17th Ed.). The Essential Guide for Writers, Editors and Publishers. The University of Chicago Press.
- Peter Coxhead, Writing a Project Report, University of Birmingham. URL: https://www.cs.bham.ac.uk/~pxc/proj/ProjectReport.pdf
- Wilhelmiina Hämäläinen. 2006. Scientific Writing for Computer Science Students, University of Joensuu. URL: http://cs.joensuu.fi/pages/whamalai/sciwri/sciwri.pdf
- William J. Rapaport (2018) How to Write (How to Prepare Technical Reports), State University of New York at Buffalo. URL: https://cse.buffalo.edu/~rapaport/howtowrite.html